

Leverage time-based data to improve recommendation system accuracy

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Abstract

This study shows that the industry is developing rapidly with the development of mobile devices, and most users can easily receive various recommendations through mobile devices. In order to maintain the quality of user experience, predicting the recommendations that current users want is the main takeaway from this study. Through our experiments, we found that making recommendations based on recent user data rather than the amount of data can improve the overall recommendation accuracy. However, we also found that the amount of data is important in terms of recall score. Further research is needed to improve both accuracy and recall scores.

Keywords: Graph Neural Network, Recommendation System, Link prediction.

1. Introduction

In order to improve the quality of these services, companies have provided services to recommend items for users to buy or watch by referring to their purchase history and viewing history, and these recommendation systems are being researched in various fields.

Representative success stories include Amazon, Netflix, Facebook, Baidu, and Alibaba, which have adopted personalized recommendation systems to increase sales and brand loyalty, differentiate themselves from competitors by meeting individual customer preferences, and apply recommendations to relevant customer experiences, new users, or new catalog items in real time, with 35% of their revenue coming from recommendations, and Netflix, which is motivated by the fact that "Netflix members lose interest after 60 to 90 seconds if they cannot decide on a movie after reviewing 10 to 20 titles". In order to improve the accuracy and personalization of these services, Netflix held a technology competition called "Netflix Prize"

and offered \$1 million to anyone who could improve the quality of Netflix's recommendation system "CineMatch" by 10%. [1].

These recommendation systems are a major source of revenue for many industries, and the development of these systems requires efficient ways to use large amounts of data.

2. Recent trends in recommendation systems

The existing recommendation system has been developed based on technologies such as Collaborative Filtering, Content-based Filtering, and Hybrid Recommendation, but these methods have the disadvantage of not showing high accuracy because they do not properly capture the interaction between data users and items, and as a way to capture such interactions better than the existing ones, research on graph-based recommendation methods that express the relationship between users and items in the form of a graph is actively being conducted, and by applying a type of graph neural network (GNN),

it is possible to express the interaction between users and items in the form of a graph, and based on this, personalized recommendations can be made more accurately. [2, 3, 4, 5, 6, 7].

In this study, GNN, which can capture the relationship between users and items better than existing methods, was used in the recommendation system, and experiments were conducted on how to use the data to predict the items that users want at the current time.

3. Experiments

The goal of this experiment is to predict "recommendation of items that the current user wants", and since the experiment requires predicting the current time, the main data attribute is time, and "MovieLens 100k", which contains temporal information about user behavior, was used for the experiment. The timestamp attribute refers to the time when the user who consumed the item left the item rating, and we sorted the data in chronological order based on this attribute and divided the data to conduct the experiment. We conducted the experiment based on the idea that recommending items consumed by recent users would be more effective in performance in order to accurately predict "recommend items that the current user wants." The test data for evaluating the model's performance was assumed to be the latest data based on the timestamp, and the same data was used for all experiments. The process of the data divided by period and the experimental results by data are visually represented as A, B, C, D, and R in Figures 1, 2, and 3, where R is the data for conducting experiments using the same number of data as A, which separates only the most recent data, and is randomly extracted from the entire training data and used as training data. The performance results of the overall experimental results show that the experimental results of A using only the most recent data with the smallest number of data have higher F1, NDCG@20, MRR@20, and Precision@20 scores than learning more data.

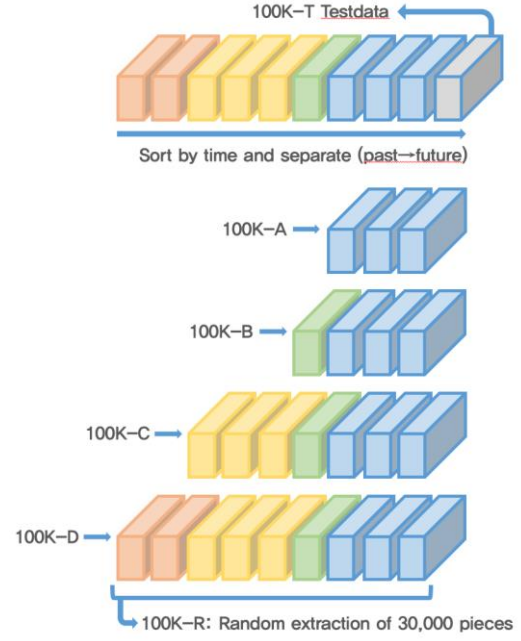


Fig. 1. MovieLens 100k Data separation schematic

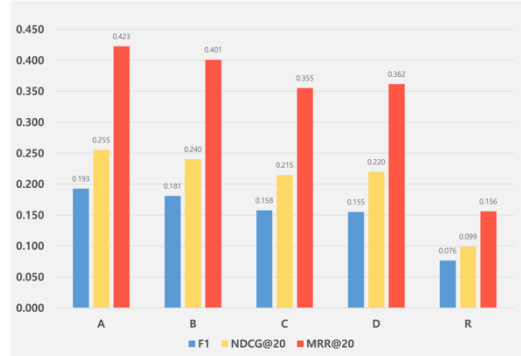


Fig. 2. Graph comparing NDCG@20 and MRR@20 performance by data

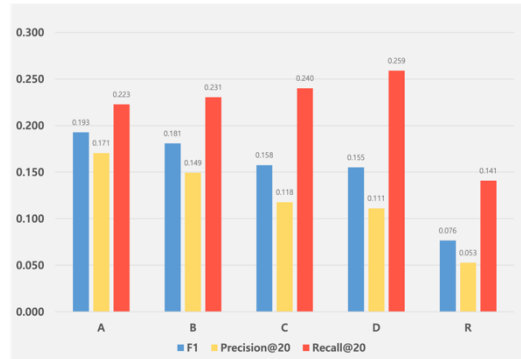


Fig. 3. Graph comparing Precision@20 and Recall@20 performance by data

4. Conclusion and future research

Most of the existing research on Collaborative Filtering, Content-based Filtering, Hybrid Recommendation, etc. is based on capturing meaningful user-item relationships by applying the entire data and new methodologies. However, in this study, we studied how to utilize data to more accurately predict the "current user's desired item recommendation" and proved through experiments that using only recent user records is more accurate. However, by learning only recent data, we found that the recall score was less accurate than learning using all data.

We believe that it is necessary to apply a methodology to improve this, and for this purpose, instead of not learning the user's past information at all, we believe that applying a method to predict the user's interests by gradually changing the user's interests based on past information is a way to improve both precision and recall scores, and we plan to conduct research for this.

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